HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed embedded C/C++ firmware for microcontrollers, focusing on driver development for real-time data acquisition and control systems.
- Designed and implemented communication protocols including **SPI** and **UART** for inter-device data exchange on embedded Linux platforms.
- Contributed to the architecture of embedded software for life safety products, ensuring modularity and adherence to system design requirements.
- Performed rigorous testing and root-cause analysis on embedded systems, resolving complex hardware-software integration issues across multiple development cycles.
- Collaborated with hardware and validation teams to integrate wireless connectivity modules, supporting **BLE** and **Wi-Fi** technologies in connected home devices.
- Participated in code reviews, enhancing code quality and maintainability, and contributed to the development of automated testing frameworks for embedded software.
- Managed production incidents for enterprise applications using **Splunk** and **Dynatrace**, ensuring adherence to SLAs and performing P1/P2 incident response.

SKILLS

Languages – C, C++, Python, SQL, Java

Embedded Systems – Microcontrollers, RTOS, Embedded Linux, ARM Cortex-M, Drivers, System Integration

Wireless Technologies – BLE, Matter, Thread, Wi-Fi, Bluetooth

Protocols – SPI, I²C, UART

Tools & Practices – Git, CI/CD, Jenkins, Docker, Debugging Tools, Hardware-Software Integration, Code Reviews, Automated Testing, Technical Documentation

Cloud & Monitoring – AWS, EC2, S3, Lambda, CloudWatch, Splunk, Dynatrace

PROJECTS

Embedded Firmware for IoT Sensor Node

- Developed firmware in **C** for an **ARM Cortex-M** microcontroller to manage sensor data collection and transmission via **BLE**.
- Implemented **SPI** drivers for sensor interfacing and a custom communication protocol for reliable data streaming to a central gateway.
- Integrated power management features to extend battery life, achieving over 6 months of continuous operation in simulated testing.

Real-Time Data Acquisition System

- Designed and built a Python-based application for real-time data ingestion and processing from embedded devices using **UART** communication.
- Developed a **Flask** API to expose processed data, supporting live monitoring dashboards and historical data retrieval.
- Configured **Docker** containers for deployment and managed the application lifecycle on **AWS EC2** instances.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)